Fiscal Year:	FY 2020	**Task Last Updated:**	FY 06/29/2020
PI Name:	Hayman née Anderson, Allison Ph.D.		
Project Title:	Performance Enhancement Through Multi-Modal Stochastic Resonance		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Boulder	**State:**	CO
Zip Code:	80309-5004	**Congressional District:**	2
Comments:	NOTE: name change to Hayman née Anderson (Ed., March 2025). PI moved to University of Colorado from Dartmouth College in early 2017.		
Project Type:	Ground	**Solicitation / Funding Source:**	2018 TRA BRASH1801: Translational Research Institute for Space Health (TRISH) Biomedical Research Advances for Space Health
Start Date:	01/01/2019	**End Date:**	12/31/2021
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	8	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2021 per E. Urquieta/TRISH (Ed., 4/14/21) NOTE: End date changed to 5/31/2021 per TRISH Task Book submission (Ed., 7/2/2020)		
Key Personnel Changes/Previous PI:	Ed. note 7/2/2020--CoInvestigators Mulavara and Bloomberg are no longer working on the project.		
COI Name (Institution):	Clark, Torin Ph.D. (University of Colorado, Boulder)		
Grant/Contract No.:	NNX16AO69A-T0402		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This project's objective is to investigate the operational benefit of brain stimulation with stochastic resonance (SR). The team will measure performance enhancement across multiple sensory modalities, in complex tasks, and over repeated use when stochastic resonance is applied through the auditory and vestibular sensory pathways. To enhance neurocognitive and task performance, stochastic resonance may be used to enhance information transfer across multiple sensory modalities, reducing the cognitive workload associated with complex tasks. SR is achieved when sub-threshold white noise is added to a perceptual channel (e.g., auditory, vestibular, visual, etc.), and resonates with the underlying primary signal. As a result, the signal's message is enhanced, allowing the human to perceive the primary signal more easily. Not only does SR increase performance on the modality to which it is applied, it also shows cross-modality benefits and improves perception in other sensory systems. For example, adding auditory white noise can improve the ability to sense touch or light. Further, by improving perception, there may be secondary benefits of SR such as improved memory, enhanced sleep, and reduced stress.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research investigates the use of SR as a noninvasive, easy to administer way to stimulate the brain and improve performance in spaceflight relevant tasks. It investigates multimodal SR through the auditory and vestibular channels without requiring bulky, difficult to wear, high power devices that cannot be used in operational environments. This research incorporates an evaluation of the long-term effects of SR in an operational spaceflight analog. SR is likely to be highly acceptable to astronauts because it is applied in a manner that is non-invasive, stimulates multi-sensory neurons in the brain through natural sensory pathways, and does not have the harmful or unpredictable outcomes seen in transcranial magnetic or current stimulation. Multi-modal SR has not previously been investigated nor applied to complex spaceflight relevant tasks. Because SR stimulates the brain and improves performance across all sensing modalities, we expect it to show improvements across a broad range of tasks. These improvements in performance achieved through SR may be critical for times of high stress, where resiliency is required and improvement in performance can make the difference, for example during vehicle launch, landing, or emergency situations. It may also show long-term benefits by reducing the cognitive demands on every-day tasks, stimulating the brain and preserving brain function. In Year 1, our research has identified novel findings (vestibular stochastic resonance for improvement in the visual perception task), confirmed findings in the literature (auditory stochastic resonance for the auditory task), and has developed novel experimental design methodologies. It has furthered our ability to objectively assess SR, adding to the literature to ensure these improvements are rigorously identified. We have begun the process of validating this work in a spaceflight relevant analog by successfully being integrated with the NASA Human Exploration Research Analog (HERA) Campaign 6. One additional area of impact not originally identified but that was added opportunistically to this study was to investigate the use of SR while performing the Cognition Test Battery, which is being done by all researchers within our cohort. We have begun the initial experimental design and will perform this assessment in Year 2. This research addresses the Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorder. This research is related to gap BMed1: we need to identify and validate countermeasures that promote individual behavioral health and performance during exploration class missions. The proposed research investigates the effects of SR on performance in spaceflight relevant tasks by increasing attention, perception, and information integration, with the potential to provide secondary benefits in memory, stress, and mood.
Task Progress:	This project's objective is to investigate the operational benefit of brain stimulation with stochastic resonance (SR). The team will measure performance enhancement across multiple sensory modalities, in complex tasks, and over repeated use when stochastic resonance is applied through the auditory and vestibular sensory pathways. To enhance neurocognitive and task performance, stochastic resonance may be used to enhance information transfer across multiple sensory modalities, reducing the cognitive workload associated with complex tasks. SR is achieved when sub-threshold white noise is added to a perceptual channel (e.g., auditory, vestibular, visual), and resonates with the underlying primary signal. As a result, the signal's message is enhanced, allowing the human to perceive the primary signal more easily. Not only does SR increase performance on the modality to which it is applied, it also shows cross-modality benefits and improves perception in other sensory systems. For example, adding auditory white noise can improve the ability to sense touch or light. Further, by improving perception, there may be secondary benefits of SR such as improved memory, enhanced sleep, and reduced stress. In Year 1, our team has made significant advancements on our stated objectives. We focused on whether brain stimulation through stochastic resonance can improve sensory perception across multiple sensing modalities. To date, we have tested an initial cohort of human subjects using vestibular stochastic resonance (electric white noise administered via galvanic vestibular stimulation to the mastoids, at 10 levels between 0.1 and 1 mA RMS), auditory stochastic resonance (white auditory noise administered in headphones at a range of levels up to 80 dB), or sham (in which headphones and galvanic vestibular electrodes are worn, but no white noise is provided). Many of these test conditions have never been investigated in prior research. As a novel finding, we found that vestibular stochastic resonance significantly improved visual perceptual thresholds at the subject-specific best current level, as compared to the sham. Furthermore, we found that subjects which had the highest visual thresholds demonstrated the largest improvements with vestibular stochastic resonance. Further, we have confirmed other research findings for in-channel auditory stochastic resonance. Data analysis and data collection is ongoing. This research has also furthered our ability to objectively assess SR, adding to the literature to ensure these improvements are rigorously identified. We have developed a unique approach to designing experiments that use stochastic resonance to influence perceptual thresholds. Specifically, we use Monte Carlo simulations to understand the effect of different experimental design manipulations (e.g., the number of trials used to estimate each threshold, the number and placement of stochastic resonance levels) on our ability to identify stochastic resonance benefits, if they exist, given various assumptions about the subjects' underlying response behavior. We have begun the process of validating this work in a spaceflight relevant analog by successfully being integrated with the NASA HERA Campaign 6. One additional area of impact not originally identified but that was added opportunistically to this study was to investigate the use of SR while performing the Cognition Test Battery, which is being done by all researchers within our Translational Research Institute for Space Health (TRISH) cohort. We have begun the initial experimental design and will perform this assessment in Year 2. Overall, our research was impacted by COVID-19, delaying our experimental deliverables. We have focused on efforts, though, on advancing experimental design for our Year 2 objective and developing computational, objectives ways to identify SR. We will use this information to further enhance our research findings as we move into Year 2.

Bibliography Type:	Description: (Last Updated: 02/18/2026)
Abstracts for Journals and Proceedings	Anderson A, Rise R, Sherman, S, Voros J, Callas M, Kryuchkov A, Stine P, Clark T. "Performance Enhancement Through Multi-Modal Stochastic Resonance." 2020 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 27-30, 2020. Abstracts. 2020 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 27-30, 2020. , Jan-2020
Abstracts for Journals and Proceedings	Rise R, Voros J, Anderson AP, Clark TK. "Using Simulation to Improve Sensory Threshold Estimation on Two-Interval Stochastic Resonance Tasks." 2020 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 27-30, 2020. Abstracts. 2020 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 27-30, 2020. , Jan-2020
Abstracts for Journals and Proceedings	Sherman S, Kryuchkov A, Stine P, Clark T, Anderson A. "Auditory Stochastic Resonance to Improve Perceptual Thresholds." 2020 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 27-30, 2020. Abstracts. 2020 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 27-30, 2020. , Jan-2020
Abstracts for Journals and Proceedings	Voros J, Callas M, Anderson AP, Clark TK. "Galvanic Vestibular Stochastic Resonance To Improve Perceptual Thresholds." 2020 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 27-30, 2020. Abstracts. 2020 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 27-30, 2020. , Jan-2020
Awards	Kryuchkov A. "Undergraduate Research Opportunity, University of Colorado, August 2019." Aug-2019
Awards	Rizkallah J. "Undergraduate Research Opportunity, University of Colorado, May 2019." May-2019
Awards	Callas M. "Undergraduate Research Opportunity, University of Colorado, August 2019." Aug-2019
Awards	Stine P. "Summer Program for Undergraduate Research, University of Colorado, May 2019." May-2019
Dissertations and Theses	Voros J. "Cross Modal Stochastic Resonance in Perceptual Thresholds with Galvanic Vestibular Stimulation." MS Thesis, The University of Colorado, Boulder, April 2020. , Apr-2020
Papers from Meeting Proceedings	Voros J, Sherman S, Rise R, Callas M, Kyruchkov A, Stine P, Rizkallah J, Anderson A, Clark TK. "Multi-Modal Stochastic Resonance to Enhance Astronaut Perceptual Performance: Experimental Design." 2020 IEEE Aerospace Conference, Big Sky, MT, March 7-14, 2020. 2020 IEEE Aerospace Conference, Big Sky, MT, March 7-14, 2020. , Mar-2020